

The Anagram Grouping Task

A take-home project · Please return within one week

THE TASK

Write a command-line program that accepts a single argument — a path to a text file containing a large list of words, one per line. The program should re-organise the words so that all anagrams of each other appear together on a single line, separated by spaces.

Usage example: `my_program data_file.txt`

EXAMPLE

INPUT

```
apple
car
cider
tar
itch
rat
cried
helicopter
arc
```

OUTPUT

```
apple
car arc
cider cried
tar rat
itch
helicopter
```

WHAT WE'RE LOOKING FOR

ALL CANDIDATES

- A working, runnable naive implementation
- Code that's easy for the whole team to read and understand

THEN, DEPENDING ON YOUR FOCUS

BACK-END FOCUS

- Thoughts on scaling when the file doesn't fit in memory
- This can be written explanation, diagrams, or code
- If your scaled solution is complex, include the naive version too

FRONT-END FOCUS

- A web, mobile, or other user interface
- Even a rough wireframe or diagram is perfectly acceptable

LANGUAGE & APPROACH

Use any language or format you feel is appropriate. You may even choose to submit descriptions or diagrams alone. There are no strict rules here — use the task as an opportunity to show us how you think.

TIME & SCOPE



We're looking for **quality over quantity** — you don't need to complete every part. Narrowing the scope and doing something well is far more valuable than rushing through everything. We'll be reviewing your submission closely and discussing it with you at the next stage, so treat it as a conversation-starter rather than a checklist.

Please aim to return your submission within **one week**. If you need more time, just let us know — that's absolutely fine.